



LATER GATOR

SPRINT 2 PRESENTATION

PREPARED FOR

CEN3031 – Introduction to Software Engineering

FALL 2025 | University of Florida

Professor Lisha Zhou

PREPARED BY

TEAM LATER GATOR:

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Carrie Ruble

Samantha Wong

Presentation Overview

The following pages provide the slides and corresponding speaker notes for Team Later Gator's *Sprint 2 Presentation*. These materials outline recently completed user stories, updates to our architecture and database, development milestones, testing workflows, and a demonstration of the features implemented during this sprint. We conclude with our goals and priorities for the next phase of development.

A recording of the full presentation can be viewed at the following unlisted YouTube link:

<https://youtu.be/q7ndh-UDIpq>.



SAM: "We are **Team Later Gator**, and today we're presenting our progress on the project. --

I'm Sam – *Project Manager*, responsible for helping define the product vision, prioritize product backlog features, and ensure the team develops feature based on the user's needs. "

OTHERS:

I'm **Madison** – *Scrum Master*, responsible for maintaining our Jira and ensuring our sprints go according to plan

I'm **Danielle** – *Developer*, responsible for ... Ensuring correct coding environment and tools and that product development best practices are followed.

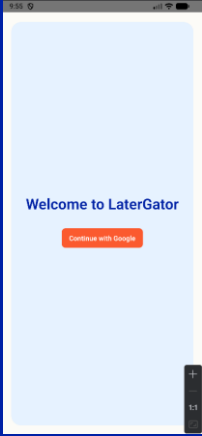
I'm **Carrie** – *Developer*, responsible for ... I work with Danielle on keeping the team well resourced for our product development.



**SPRINT 1
ACCOMPLISHMENTS**

PROJECT UPDATE

- 1. SETUP THE FULL DEVELOPMENT ENVIRONMENT**
 - Used Android Studio and Kotlin
 - Pure Android Native app
 - Organized project structure
- 2. RESOLVED MAJOR TOOL ISSUES**
 - Build issues with Room and React Native
 - Tools did not communicate as expected
 - Pivoted to simpler structure
 - Firebase – login in screen error
 - Updated SHA 1 fingerprint
 - Different errors while building environment in different team member computers



SAM:

We set up the full development environment and created the original database structure using Room, Android Studio, and Kotlin. However, because the tools were not communicating as expected and Room caused repeated build issues, we pivoted to a simpler structure without Room and react native. We now have a **Pure android native app**. We also ran into Firebase login problems and couldn't get past the authentication screen until we updated the SHA-1 fingerprint in our Firebase project and configuration file. On top of that, different team members were experiencing different build errors in their environments, which we resolved through troubleshooting.

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Slide 3

PROJECT UPDATE

3. BUILT APP FEATURES

- Created UI for home screen, setting screen, login

4. CREATED EARLY TEST CASES



**SPRINT 1
ACCOMPLISHMENTS**

Welcome to Leter Galur, Dani

Periodio Timer

Settings

Sign Out

Settings Screen

Username
Dani

Update

Usage Access Permission Status: Not Granted

Grant Permission

Tracked Apps Manage

Gmail 28 min

Maps No limit

Back to Home

SAM:

Despite these challenges, we were able to develop the Home, Login, and Settings screens and create test cases to verify that these features worked as intended.

NEXT SLIDE

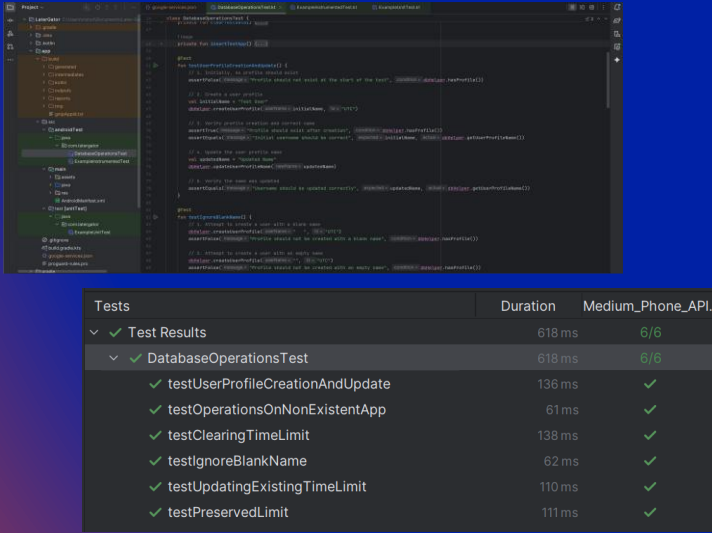
CURRENT TESTING



**Android Studio
TEST SUITE**



FIREBASE



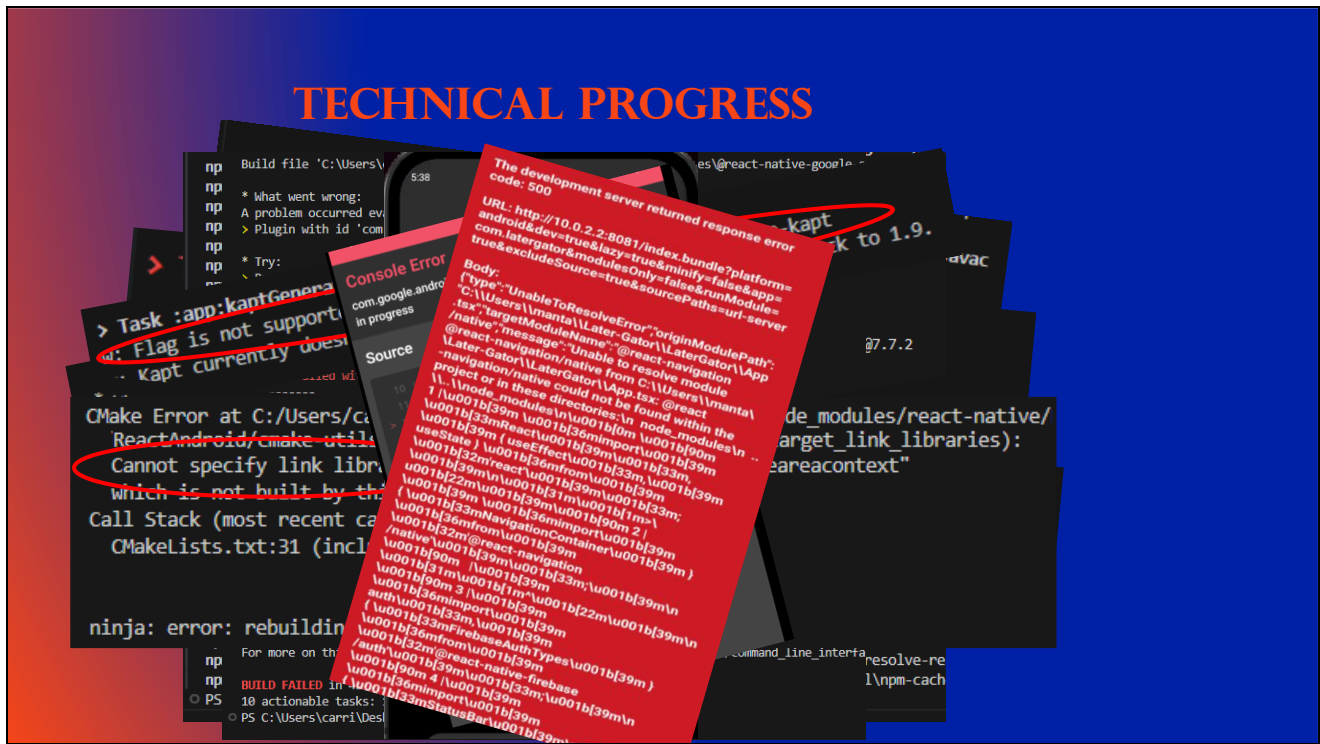
Tests	Duration	Medium_Phone_API
✓ Test Results	618 ms	6/6
✓ DatabaseOperationsTest	618 ms	6/6
✓ testUserProfileCreationAndUpdate	136 ms	✓
✓ testOperationsOnNonExistentApp	61 ms	✓
✓ testClearingTimeLimit	138 ms	✓
✓ testIgnoreBlankName	62 ms	✓
✓ testUpdatingExistingTimeLimit	110 ms	✓
✓ testPreservedLimit	111 ms	✓

MADISON:

This had us switch to building our entire project in Android Studio and using their test suite to validate and test our build. With Firebase fully implemented, we built our test cases around validating that user set up is initiated and all fields correct write to the database and update when appropriate. We also validate that app usage limits are set and updated and persist throughout the app. When we implement more elements in our next sprint, validating test cases will be a part of success criteria.

Carrie will discuss our technical progress through this sprint.

NEXT SLIDE



CARRIE:

Thanks, Madison!

[SCROLL THROUGH IMAGES AS READING BELOW]

Over this sprint, most of our technical progress centered around getting our foundational tools to actually work together.

The decision was made early-on to build our Android prototype using React Native, and Room for our on-device database. In theory, that seemed straightforward, but in practice, it became the biggest technical obstacle of the entire sprint. – And hopefully, of our whole project!

Danielle flagged a serious problem: even with all our dependencies installed, she kept getting this error you see on the screen. A file that should have been system-generated, wasn't.

Every attempt to resolve this error resulted in large, conflicting Gradle and Kotlin errors. At first we thought it was something small, maybe a missing plugin

or one outdated version.

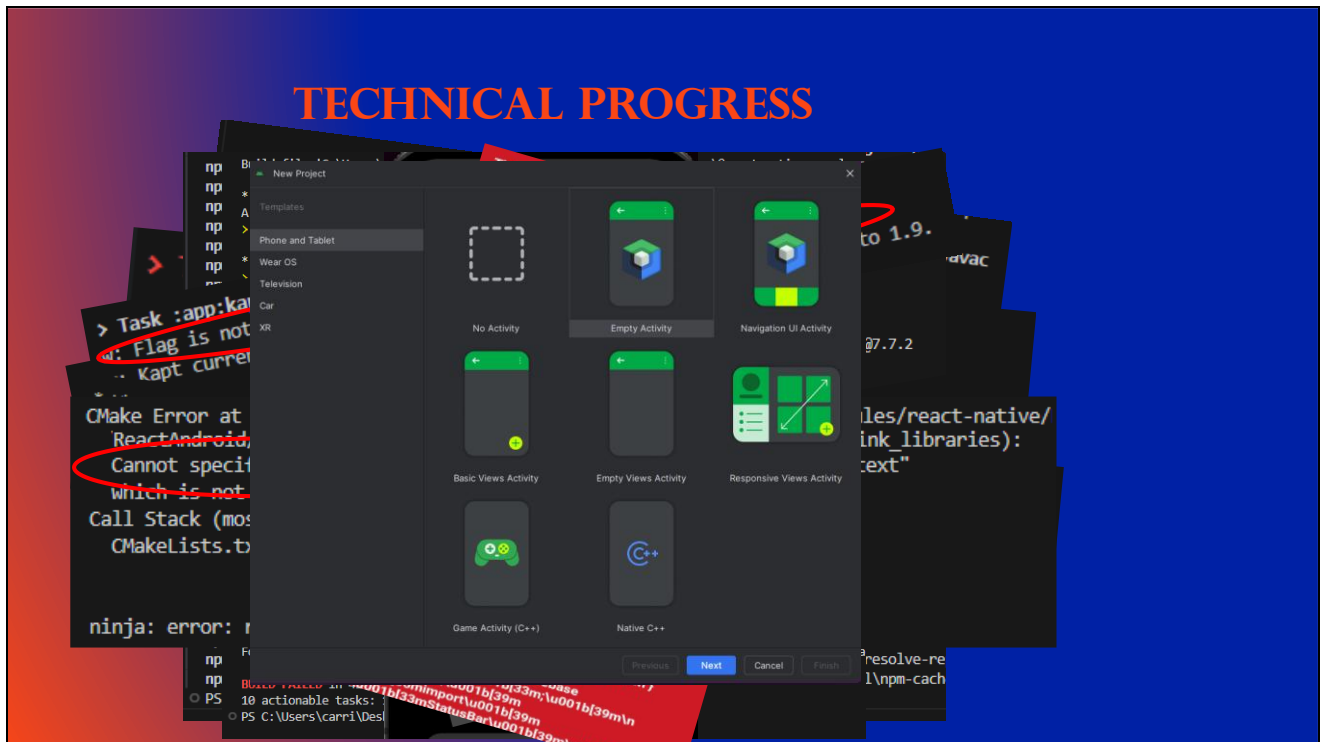
But as we dug deeper, the failures pointed to something much bigger.

The combination of React Native, Firebase, Kotlin, KSP or KAPT, and Room inside a single project introduced a cascade of compatibility issues.

As a team, we spent hours comparing logs, version numbers, and plugin configurations. We tried several different approaches: upgrading Gradle, downgrading Kotlin, swapping KAPT for KSP, adjusting annotation processors, restructuring directories, and even rewriting parts of the Room layer. Each time we fixed one error, another appeared immediately afterward. The errors often contradicted each other, fixing one broke something else. And because none of us on the team had mobile development experience, there was a steep learning curve simply figuring out what the errors *meant*.

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CARRIE:

Eventually, it became clear that the integration itself—not our code—was the root problem.

The React Native template was not designed for the level of native Room integration we were attempting.

We realized that if we continued fighting Room, we would risk losing the entire sprint or even failing to deliver a working prototype.

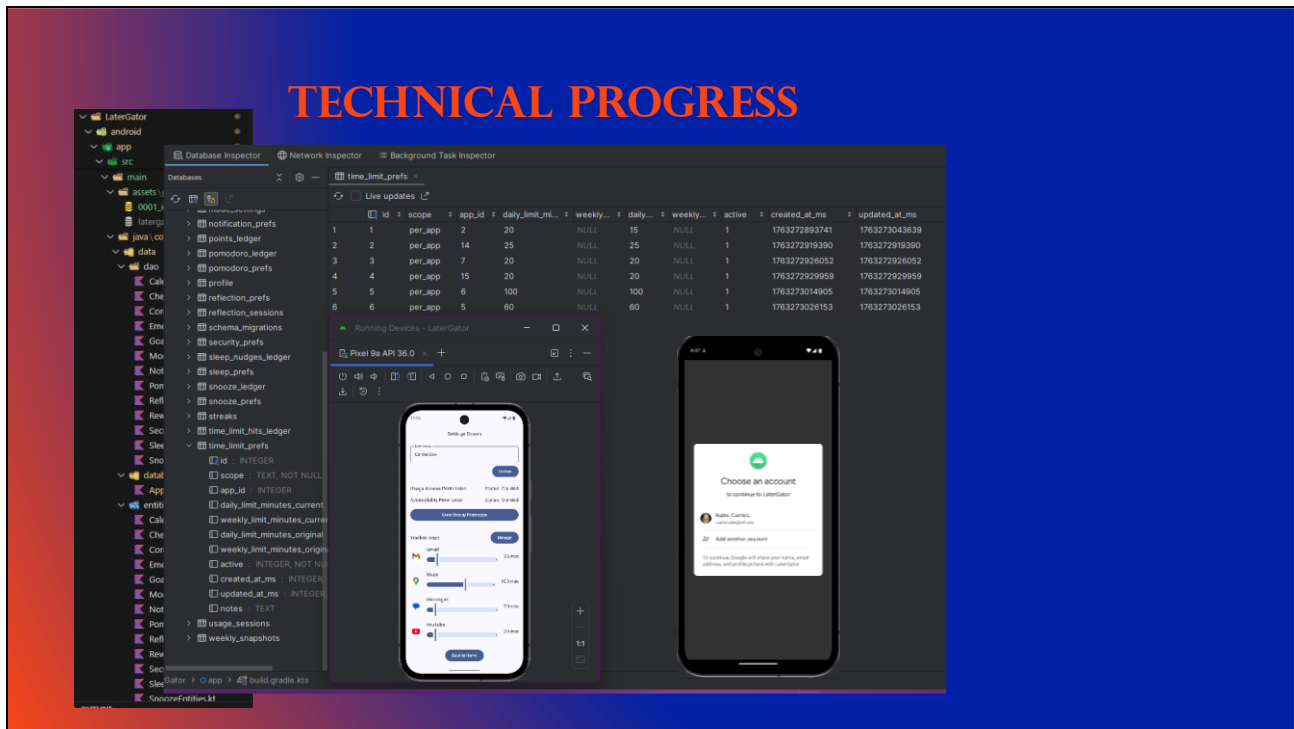
CLICK

So, we made a strategic pivot.

CLICK

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Slide 8



CARRIE:

Instead of relying on Room, we rebuilt our data layer using a simpler, more reliable method: a raw SQLite database stored directly in the app's assets folder.

On install, the app copies that database onto the device and uses standard SQLite queries to read from it and write to it.

Once we removed Room and cleaned the project, everything fell into place.

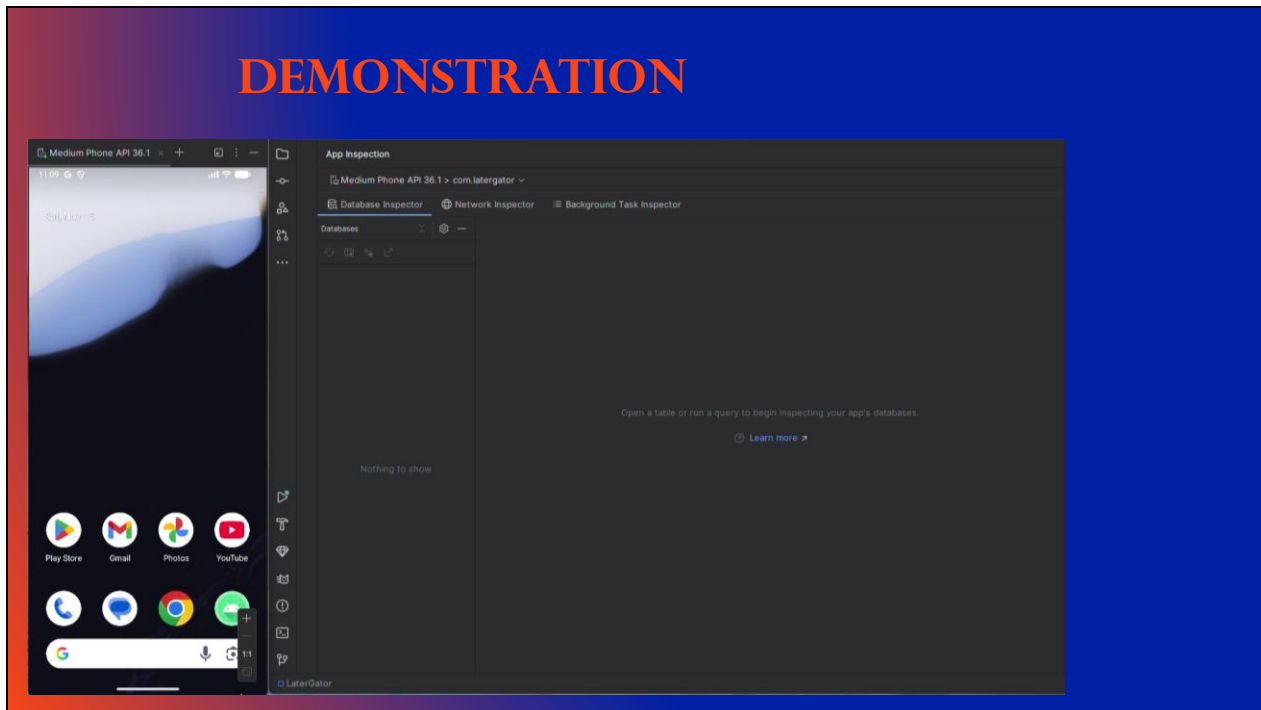
CLICK

The app compiled successfully, Firebase Auth remained fully functional, and our SQLite database was detected and readable on the emulator. Seeing the project finally launch, and seeing real data coming from our database, felt like a turning point.

And next, Danielle is going to show you that we were still able to make practical progress toward our prototype during this sprint.

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DANIELLE:

Thanks Carrie! I'll now walk through and demonstrate the progress that the team has made, despite our setbacks.

- Later Gator uses federated authentication, having users sign-in with Google
- Prompts user to enter a username so that a profile can be created (username and timezone)
- Then if the user wants to change their name they can do so here.
 - The user can't enter a blank text box as their name
- New input is updated in the database can be seen in the greeting on the home screen
- Later Gator will need permission to access app usage stats, so there is a button allowing the user to do that
- Now for selecting apps to track:
 - here are all the apps on the device
 - Users select the ones they wish to track, and they will appear on the main settings screen
 - Apps database table will show they are being tracked

- When apps are first selected, they do not have a time limit set (therefore don't immediately show in time_limit_prefs table)
- Once the user moves the slider, a row is inserted into the table to store the user's time limit for that app
- When the user modifies the slider value, the current limit is adjusted accordingly
- When an app is unselected, it is removed from the settings page and is no longer marked as tracked
- If a previously selected app were to be re-selected, the limit will return to whatever it was last set at since it is just inactivated when deselected, not removed from the time_limit_prefs table
- This button takes the user to the Pomodoro Study screen where they'll be able to initiate a Pomodoro study session
- And finally, there is a button to sign the user out of the app

Now over to Madison for what our next Sprint holds

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LAST SPRINT

Item ID	Description	Time Estimate	Due Date	Story Points
SP1-11	Frontend Login Page	10 DO	Oct 19	3
SP1-13	Frontend Settings Menu	10 DO	Oct 19	3
SP1-14	Frontend Home Screen	10 DO	Oct 19	3
SP1-15	Frontend Popup Notifications	10 DO	Oct 19	1
SP1-16	Frontend Pomodoro Study Session	10 DO	Oct 19	2
SP1-17	Frontend Reports Screen	10 DO	Oct 19	5
SP1-18	Frontend System Notifications	10 DO	Oct 19	2
SP1-19	Backend Login Page	10 DO	Oct 19	3
SP1-21	Backend Settings Menu	10 DO	Oct 19	8
SP1-23	Backend Popup Notifications	10 DO	Oct 19	2
SP1-24	Backend Pomodoro Study Session	10 DO	Oct 19	2
SP1-25	Backend Reports Screen	10 DO	Oct 19	5
SP1-26	Backend System Notifications	10 DO	Oct 19	2

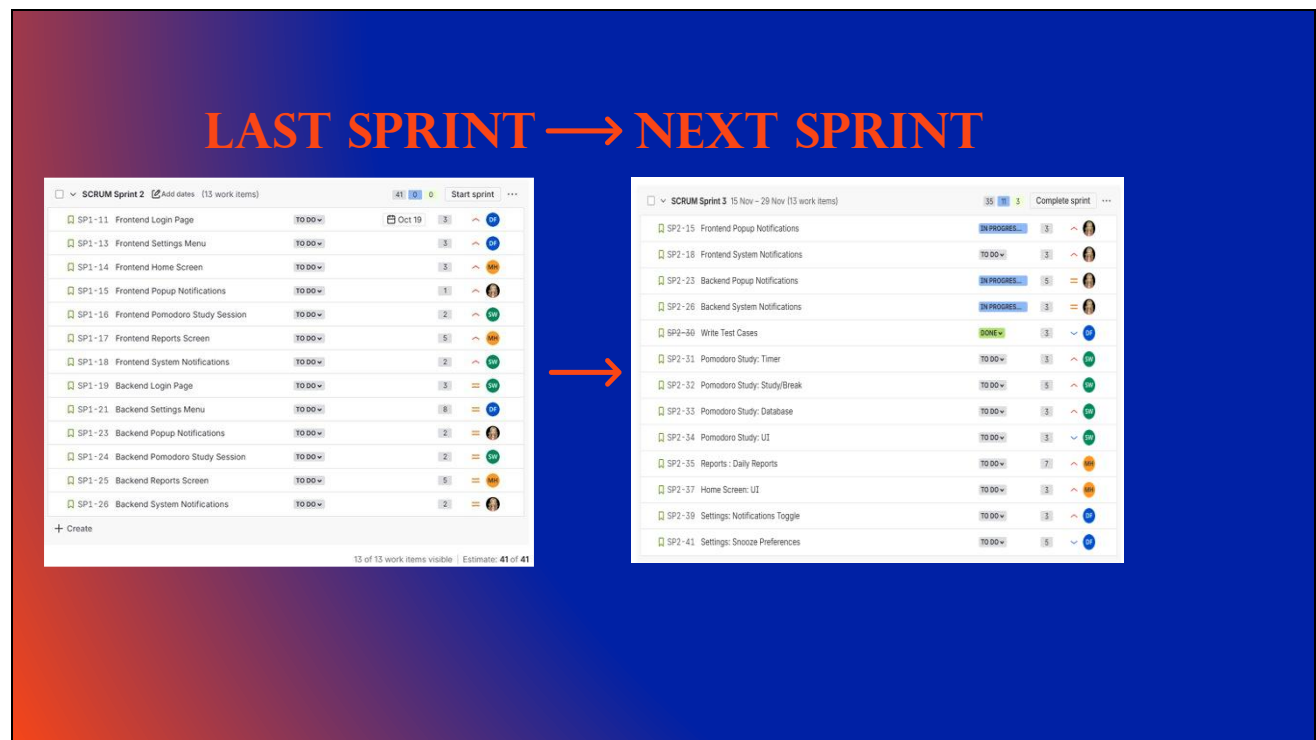
- ITEM DESCRIPTIONS TOO VAGUE
- NO SUCCESS CRITERIA
- STORY POINT ESTIMATIONS TOO LOW
- NO WIGGLE ROOM FOR ERRORS AND UNEXPECTED CIRCUMSTANCES
- HOW TO REPRIORITIZE AFTER SET BACKS?

MADISON:

After completing our Sprint Retrospective, we realized our expectations for what we could accomplish in this Sprint were too optimistic. We found that our Item descriptions were too broad and vague. Tasks needed to be broken down further into more manageable chunks with better defined success criteria. Our Story Point Estimations were found to be far too low. We gave ourselves very little wiggle room for unexpected errors and technical difficulties. When we encountered issues in our development, it became impossible to complete all the cards in our sprint and we needed to reprioritize what features could reasonably be completed within the remaining time with all of our setbacks.

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Slide 11



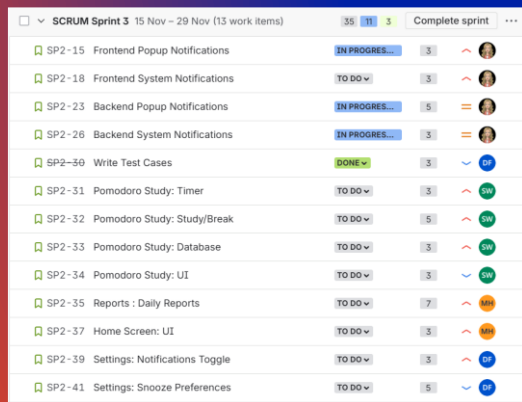
MADISON:

Keeping all of these things in mind, we restructured our backlog items to better fit our needs in this next sprint

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Slide 12

NEXT SPRINT



SCRUM Sprint 3 15 Nov - 29 Nov (13 work items)		35	11	3	Complete sprint	...
SP2-15	Frontend Popup Notifications	IN PROGRESS	3			
SP2-18	Frontend System Notifications	TO DO	3			
SP2-23	Backend Popup Notifications	IN PROGRESS	5			
SP2-26	Backend System Notifications	IN PROGRESS	3			
SP2-30	Write Test Cases	DONE	3			
SP2-31	Pomodoro Study: Timer	TO DO	3			
SP2-32	Pomodoro Study: Study/Break	TO DO	5			
SP2-33	Pomodoro Study: Database	TO DO	3			
SP2-34	Pomodoro Study: UI	TO DO	3			
SP2-35	Reports : Daily Reports	TO DO	7			
SP2-37	Home Screen: UI	TO DO	3			
SP2-39	Settings: Notifications Toggle	TO DO	3			
SP2-41	Settings: Snooze Preferences	TO DO	5			

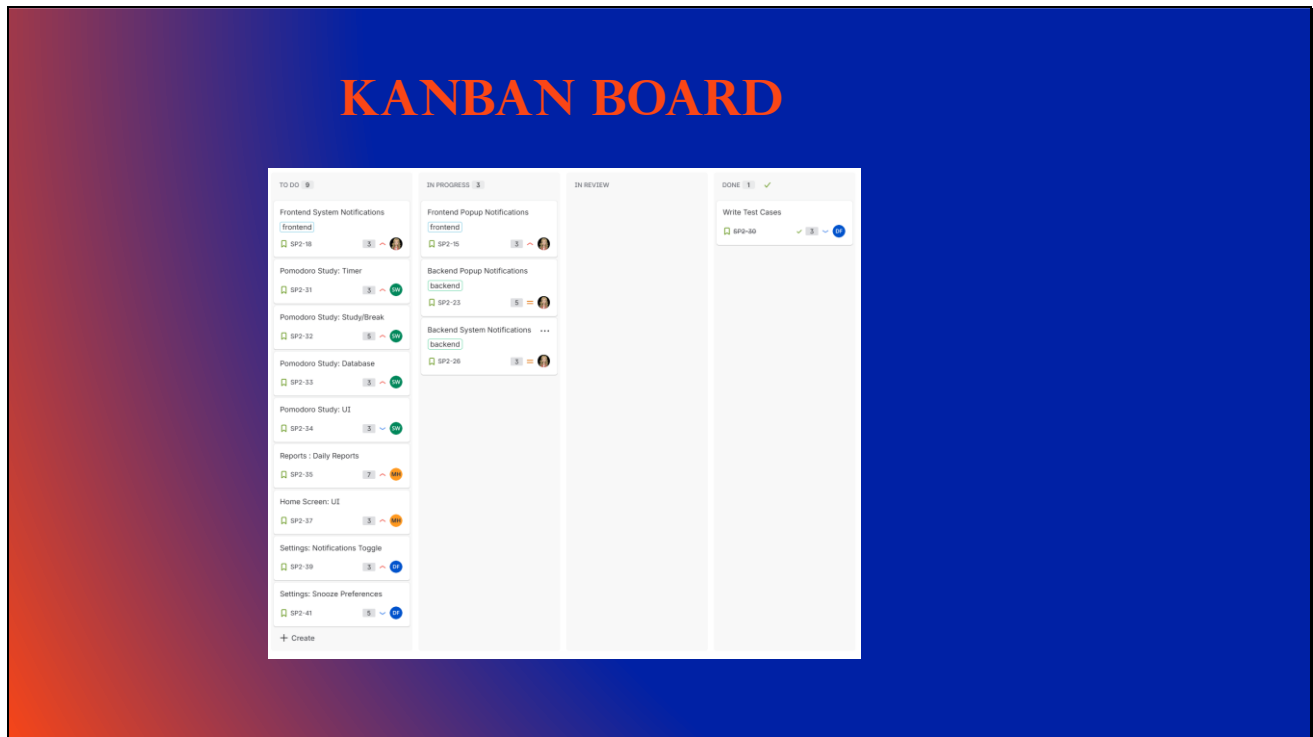
- MORE REALISTIC AND MANAGEABLE
- SUCCESS CRITERIA IN DESCRIPTIONS
- ADDED STORY POINTS TO ITEMS TO ALLOW FOR ISSUES
- REDISTRIBUTED TO TEAM MEMBERS BASED ON FEATURE

MADISON:

Our Sprint 3 card items now are more realistic and manageable. Success criteria is well defined and written in each card's description. Story point values were increased to allow for potential development issues, and cards were redistributed to team members based on deliverable features. Paired Programming on each of these issues is to be expected as it's been very beneficial in our development.

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Slide 13



MADISON:

As seen on our Kanban board, we had some elements in production from last sprint so those stayed there when moving to this new sprint. Hopefully with all of the knowledge of last sprint and many major issues worked out, this sprint will be more successful.

Onto Sam to wrap up what we learned and accomplished this sprint.

NEXT SLIDE

WRAP-UP

LESSONS LEARNED

- **UNDERESTIMATED TIME FOR EACH PRODUCT BACK LOG ITEM**
 - Breakdown task into smaller part and allocate more time
 - Build in time for potential problems
- **PAIRED PROGRAMMING**
 - Working together helped solved problems faster
 - Communicating issues on slack allowed the team to be updated on current problems

MOVING FORWARD

- **DEVELOPING FEATURES**
- **ALLOTING MORE TIME FOR BACKLOG ITEMS**

SAM:

This experience, although extremely challenging, gave us valuable insight into what it actually means to adapt during development. As Madison mentioned, we underestimated time for development. As a team, we learned how important it is to recognize when a tool is slowing development instead of supporting it. We learned how to evaluate trade-offs quickly, how to balance ideal architecture against real constraints, and how to pivot as a team without losing momentum. And most importantly, we learned how to collaborate under pressure and support each other when things weren't working.

The result is that by the end of this sprint, we have a stable, buildable project that launches cleanly, connects with Firebase, and uses a working on-device SQLite database; exactly what we needed to move forward.